

BIOS 794: Unsupervised Learning: Clustering

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Clustering

- ▶ The goal is to identify *natural groupings* in the data without relying on an outcome variable.
- ▶ Clustering is an unsupervised learning technique that is one of the most natural exploratory data analysis (EDA) methods.
- ▶ Examples in public health:
 - ▶ Grouping patients by symptom patterns or biomarker profiles.
 - ▶ Identifying communities with similar socioeconomic or health characteristics.
 - ▶ Categorizing hospitals based on performance indicators.

Clustering

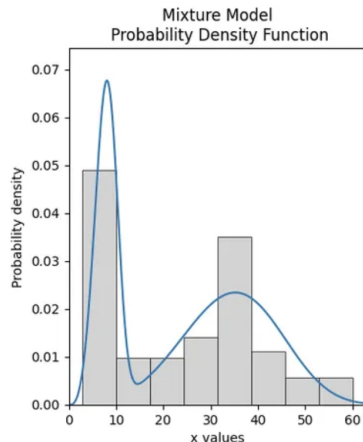
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- ▶ Clustering is an unsupervised learning technique that is one of the most natural exploratory data analysis (EDA) methods.
- ▶ Examples in public health:
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 - ▶ Identifying communities with similar socioeconomic or health characteristics.
 - ▶ Categorizing hospitals based on performance indicators.
- ▶ **Differences from classification**
 - ▶ we don't know the number of classes
 - ▶ we don't know anybody's class (no ground truth)
 - ▶ the goal is more exploratory and less predictive
 - ▶ we can cluster observations (i.e., the Y_i 's), the predictor variables (i.e., the X_i 's), or both.

Definition

- ▶ Defining what a cluster is can be difficult and seems subjective.
- ▶ Statistically, we think of clusters as a mixture distribution:

$$p(\mathbf{X}|\Phi) = \sum_{k=1}^K I(\mathbf{X} \in \mathcal{C}_k) p_k(\mathbf{X}|\phi_k).$$

- ▶ For example, the $f_k(\mathbf{X}|\phi_k)$ could be, for example, different multivariate normal distributions.



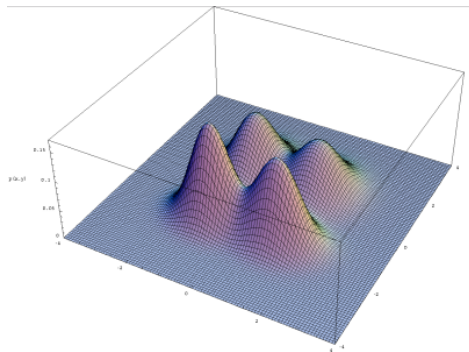


Figure: Multivariate Mixture of Gaussian distributions.

Dissimilarity

- ▶ Many clustering algorithms rely on the dissimilarity (or distance) metric used.
- ▶ Let's think of $\mathbf{x}_i$ and $\mathbf{x}_j$ as the biomarker profiles (e.g., immune response measures) between patients i and j .
- ▶ The dissimilarity between $\mathbf{x}_i$ and $\mathbf{x}_j$ has the following properties
 1. $d(\mathbf{x}_i, \mathbf{x}_j) \geq 0$
 2. $d(\mathbf{x}_i, \mathbf{x}_j) = 0 \Leftrightarrow \mathbf{x}_i = \mathbf{x}_j$
 3. $d(\mathbf{x}_i, \mathbf{x}_j) = d(\mathbf{x}_j, \mathbf{x}_i)$

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- ▶ **Examples**
 - ▶ Euclidean: $d(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{\sum_{k=1}^p (x_{ik} - x_{jk})^2}$
 - ▶ Manhattan: $d(\mathbf{x}_i, \mathbf{x}_j) = \sum_{k=1}^p |x_{ik} - x_{jk}|$
 - ▶ 1-correlation: $d(\mathbf{x}_i, \mathbf{x}_j) = 1 - \rho_{ij}$

K-means: Concept

- ▶ One of the most widely used clustering algorithms.
- ▶ Objective: divide data into K groups (clusters) that minimize within-cluster variability (i.e., the within-cluster distances).
- ▶ Each observation is assigned to the cluster with the *nearest* mean (centroid).

Example:

- ▶ Suppose we have county-level data on obesity rate, smoking rate, and median income.
- ▶ K-means can group counties into clusters that share similar health risk profiles.

K-means: Objective Function

- ▶ K-means minimizes the total within-cluster sum of squares:

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

where μ_k is the mean of cluster C_k .

- ▶ Algorithm steps:
 1. Choose K (number of clusters).
 2. Randomly K cluster centers (could be K random data points).
 3. Compute cluster means.
 4. Reassign points to the nearest mean.
 5. Repeat until assignments stop changing.

Choosing K

- ▶ For all of the clustering algorithms, we need to specify K , the number of clusters in the given dataset.
- ▶ Common methods:
 - ▶ **Elbow method:** plot total within-cluster sum of squares vs K .
 - ▶ **Gap Statistic:** measures when the improvement in fit is not better than what is expected by chance.
- ▶ In public health, interpretability also matters—choose K that makes sense in context (e.g., 3-5 community profiles).

Elbow method

- ▶ Let W_k be equal to the within-cluster distances for a given algorithm when the number of clusters is k .
- ▶ Let k^* be the true number of clusters.
- ▶ We'd expect that

$$\{W_k - W_{k+1} | k < k^*\} \gg \{W_k - W_{k+1} | k \geq k^*\}$$

.

- ▶ The difference in consecutive W_k values will be larger when $k < k^*$.
- ▶ An estimate of K is then obtained by identifying a “kink” in the plot of W_K as a function of K .

Gap Statistic

- ▶ A method to determine if $W_k - W_{k+1}$ is “large” or “small” is the Gap Statistic.
- ▶ This is a Monte Carlo method, multiple datasets with no clusters are generated with a similar spread to the variables in the original data.
- ▶ **Random Gap Values:** the $W_k - W_{k+1}$ values from these *random* datasets
- ▶ Use the Random Gap Values to judge if the observed values are larger than what would be expected by chance.

Gaussian Mixture Models (GMMs)

- ▶ K-means assumes spherical clusters of similar size.
- ▶ Gaussian Mixture Models (GMMs) assume each cluster follows a multivariate normal distribution.
- ▶ Each observation has a **probability of belonging** to each cluster.

Example:

- ▶ In a nutrition dataset, GMMs can identify overlapping groups such as “mostly healthy eaters” vs “moderate” vs “unhealthy.”

Expectation-Maximization (EM) Algorithm

- ▶ Used to estimate GMM parameters.
- ▶ Alternates between:
 - ▶ **E-step:** Estimate cluster membership probabilities.
 - ▶ **M-step:** Update cluster means and covariances based on probabilities.
- ▶ Stops when parameter changes are very small.
- ▶ Output includes both the most likely cluster assignment and the uncertainty of membership.

Partitioning Around Medoids (PAM)

- ▶ Similar to K-means but uses **medoids** (actual observations) instead of means.
- ▶ More robust to outliers because medoids minimize total distance within clusters.
- ▶ Works with any distance measure (not just Euclidean).

Example:

- ▶ PAM could be used to cluster hospitals using a mix of numeric and categorical performance indicators.

Hierarchical Clustering

- ▶ Hierarchical Clustering is a procedure that results in different degrees of clustering.
- ▶ There are two ways to do this:
 1. **Agglomerative:** Bottom-up: each point starts as its own cluster and ends up in one cluster.
 2. **Divisive:** Top-down, everyone starts as one cluster and ends up in their own cluster.
- ▶ These have similarities to forward-backward selection methods.
- ▶ For either method, observations are joined (or separated) by how dissimilar they are.

Hierarchical Clustering

Algorithm for agglomerative clustering:

1. Everyone starts in their own cluster.
2. Find the smallest d_{ij} , say d_{IJ} . Put I and J in the same cluster.

Now we have to recompute the dissimilarities for the new group IJ .

3. Compute dissimilarities $d_{IJ,k}$ for all $k \neq I$ or J . Three options for doing this:
 - ▶ **single**: $d_{IJ,k} = \min(d_{Ik}, d_{Jk})$
 - ▶ **complete**: $d_{IJ,k} = \max(d_{Ik}, d_{Jk})$
 - ▶ **average**: $d_{IJ,k} = \text{avg}(d_{Ik}, d_{Jk})$
4. Repeat steps 2–3 $n - 1$ times.

Hierarchical Clustering

- ▶ Single is more likely to cluster an observation with a group of observations.
- ▶ Complete is more likely to create new clusters.
- ▶ The algorithm for divisive clustering begins with the entire data set as a single cluster, and recursively divides one of the existing clusters into two daughter clusters at each iteration in a top-down fashion.
- ▶ We won't discuss how this works, and divisive clustering is less common than agglomerative.

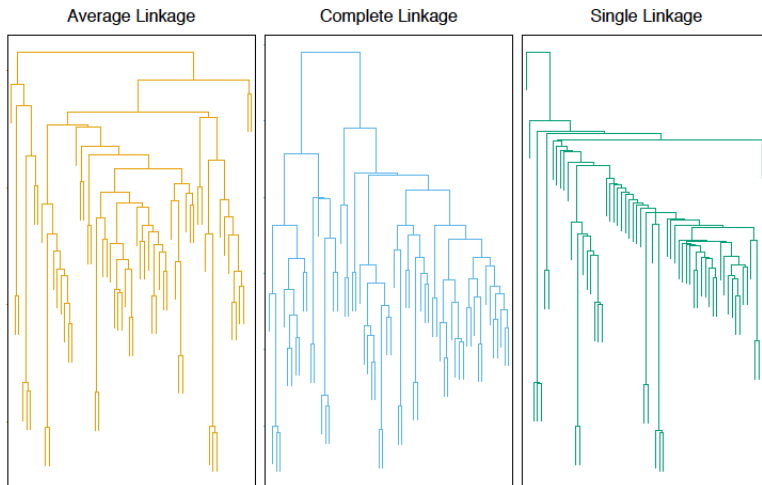


Figure: From page 524 in the online version of ESL.

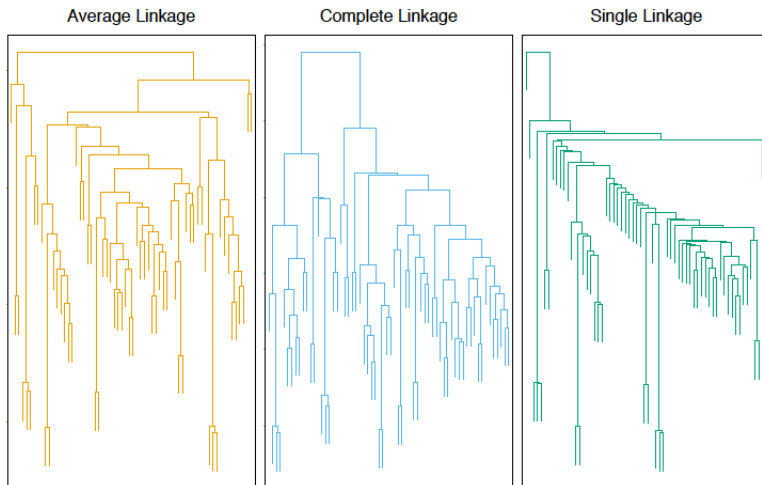


Figure: From page 524 in the online version of ESL.

Using Clustering in Downstream Models

- ▶ In practice, clustering results are often used as predictors or group indicators in later analyses.
- ▶ **Examples:**
 - ▶ Use cluster membership as a covariate in a regression model for predicting disease risk.
 - ▶ Compare outcomes (e.g., mortality, quality of life) across clusters identified from biomarker or behavior data.
 - ▶ Use clusters to define subgroups for targeted interventions or policy evaluation.
- ▶ This approach allows us to connect unsupervised patterns with meaningful outcomes.
- ▶ **However**, it changes the scientific question: we are no longer just describing structure, but testing hypotheses *about the clusters themselves*.

Pitfalls and Recommendations

- ▶ **Main Concern:** The same data are used to *create* and *test* clusters.
- ▶ This can lead to:
 - ▶ Overstated associations (data snooping bias)
 - ▶ Invalid p-values or confidence intervals
 - ▶ Misleading conclusions about “differences” between clusters
- ▶ **Good Practices:**
 1. If possible, use an independent dataset to test relationships between clusters and outcomes.
 2. Use *cross-validation* or bootstrapping to assess the stability of the clusters.
 3. Treat clustering results as *exploratory*—use them to generate, not confirm, hypotheses.
 4. Report how sensitive your downstream results are to the chosen number of clusters and method.

When to Use Each Method

- ▶ **K-means:** fast, works well for continuous data with spherical clusters.
- ▶ **GMM (model-based):** captures overlapping and elliptical clusters.
- ▶ **PAM:** robust to outliers, works with mixed data types.
- ▶ **Hierarchical:** good for exploring structure or visualizing relationships.

Key takeaway: Always pair clustering with interpretation and domain expertise—machine learning finds structure, but you decide if it makes sense.

Summary

- ▶ Clustering is essential for exploratory analysis in public health and many other fields.
- ▶ Always pre-process data: scale variables, handle outliers, and ensure numeric format.
- ▶ Use visualization to interpret results (e.g., scatter plots, silhouette plots, dendrograms).
- ▶ Combine statistical performance and interpretability when deciding the number of clusters.
- ▶